Learning-Based Demand-Supply-Coupled Charging Station Location Problem for Electric Vehicle Demand Management

Yang Song[1], Xianbiao Hu[2*]

[1]PhD candidate. Department of Civil and Environmental Engineering. The Pennsylvania State University, University Park, PA, 16802-1408. Email: ysong@psu.edu

[2]Assistant Professor. Department of Civil and Environmental Engineering. The Pennsylvania State University, University Park, PA, 16802-1408. Email: xbhu@psu.edu

ABSTRACT

We present a learning-based, demand-supply-coupled optimization model for the charging station location problem (CSLP), aiming to integrate the concept of electric vehicle (EV) charging demand management into the planning of charging infrastructures. In stage one, a gradient boosting-based learning model is developed to predict the charging demand of a charging station based on 15 defined features. Next, in stage two, a demand-supply-coupled CSLP model is developed to optimize the total charging usage rates of both existing and newly selected charging stations. We design a gradient-based stochastic spatial search algorithm to solve the proposed model. A case study with 6-year charging event data from Kansas City Missouri is performed. Results show that the proposed method can generate satisfactory charging demand predictions, and can increase charging usage rates by 14%, outperforming two benchmark approaches. The results of this research are poised to guide agencies in identifying optimal locations for new charging stations.

Highlights:

1. Predict explicit charging demand in charging station location problem without EV O-D demand matrix.
2. Model the demand-supply-coupled relationship in EV charging infrastructure design.
3. Outperform two benchmark models with a higher usage rate at charging stations.

Keywords: electric vehicle; charging station location problem; demand management; demand-supply interactions;

1. INTRODUCTION

Global climate change has prompted government agencies to champion environmentally friendly policies that emphasize sustainable and cleaner energy consumption. As transportation activities contribute to over 33% of total greenhouse gas emissions—with internal combustion engine (ICE) vehicles accounting for around 75% of these transport-related emissions (Muratori et al. 2023)—there's a pressing need to replace ICE vehicles with electric vehicles (EVs) that are powered by electric motors that emit no carbon-based emissions. However, despite the benefits of EVs, their market penetration remains relatively low. This is largely due to the technical challenges like a short driving range and extended recharging time (Kchaou-Boujelben 2021). Thus, it is vitally important to expand recharging infrastructures with a theoretically sound methodology, which can further support and incentivize the rising adoption of EVs.

When compared with those driving ICE vehicles, the travel behavior of EV travelers exhibits its unique characteristics. Range anxiety, defined as the fear that an EV will not have enough battery power to reach its destination, sets a challenging obstacle to the widespread application of EVs and has been shown to impact a traveler's route choice and trip chaining behavior (Wang et al. 2016, Yang et al. 2016). Meaning, when EV drivers plan their travels from a predetermined origin to a destination location, the availability of charging infrastructure is important information that impacts the decision of which route they should travel on. Further, research also indicated that EV travelers' destination choice behavior may differ, depending on the EV charger availability. For example, It was found by Tuttle and Kockelman (2012) that EV drivers are more likely to choose a destination offering public charging facilities, and thus businesses such as hotels and plaza have been installing charging stations (CSs) as their amenities (Nansai et al. 2001), with the hope of attracting customers than their competitors who do not offer

these facilities. These research studies suggested that the offering of EV charging infrastructure—as a new dimension of supply that is in addition to physical roadways and traffic control devices—presents an opportunity for system-level coordination to redistribute vehicular demand in the transportation network and holds promising prospects for demand management.

The optimal planning of charging infrastructure, termed charging station location problem (CSLP), is commonly defined as the selection of optimal locations for charging stations by service providers, so that the recharging demand of EV drivers can be satisfied to the maximum extent under the limitation of the investment budget (Kchaou-Boujelben 2021). Abundant studies on CSLP exist and as reviewed by Kchaou-Boujelben (2021), can be categorized into flow-based and node-based approaches. Flow-based models (Hodgson 1990, Kuby and Lim 2005, Wang and Lin 2009, Capar et al. 2013, Yıldız et al. 2016) typically represent charging demand as a series of origin–destination (O-D) trips. In this case, DC fast charging stations are strategically positioned in a manner that ensures the distance between consecutive stations remains within the driving range. This arrangement enables drivers to complete their trip from the origin to the destination and return without experiencing a battery depletion. Following such design principles, the target is frequently formulated to optimize a surrogate system measurement, such as maximizing the flow captured on the network or minimizing the system cost. Node-based models (Klose and Drexl 2005, Stephens-Romero et al. 2010, Cavadas et al. 2015, He et al. 2016, Huang et al. 2016, Cui et al. 2019), as another popular approach, assumes that charging demand in a road network is independent of specific EV trips and instead occurs at various nodes throughout the network. Following such an assumption, the node-based demand typically stems from a group of EV drivers who prefer to recharge their vehicles at a station conveniently situated near their home, workplace, or other points of interests (POIs) like plaza, and the charging demand is commonly represented

by the size of CS service area. For this type of research, the objective is to maximize the area that nodes could cover where CSs are constructed, which is defined as the area within an arbitrary short walking distance or driving time from the CS location.

Two problems arise from these CSLP modeling approaches. First, a problematic assumption is that a known O-D demand matrix of EV drivers is available for the purpose of developing and calibrating the traffic assignment model. Following the transportation planning process, obtaining an accurate EV O-D demand requires knowledge of EV ownership at each traffic analysis zone (TAZ), such as the (approximate) home address of each EV driver, and their associated mobility patterns, such as the number of trips they make in a single day and the destinations. Such information, however, is usually not available to the planners. Additionally, the above-mentioned research also suggested that an EV driver's mobility choice decisions, especially the destination choices, are influenced by the availabilities of charging infrastructure, which indicates a demand-supply coupling relationship, i.e., the O-D demand matrix may be different with a new CSLP solution. The difficulty of obtaining an accurate EV O-D demand matrix, and the variability of O-D demand due to the coupled demand-supply relationship, significantly undermine the applications of both flow-based and node-based models in the real world.

The second issue associated with most CSLP models is the lack of explicit charging demand measurements of a charging station, such as the number of charging events per day. These explicit measurements are crucial for stakeholders' decision-making in the real world, as they provide information on the actual usage rate of the charging infrastructure. For example, these measurements are essential for determining the number of chargers needed at a station, predicting the revenues that a charger may generate, and estimating power grid load based on the charging events. Unfortunately, the surrogate optimization targets used in the CSLP modeling approaches

that are summarized above, such as maximizing the captured flow on the links or maximizing the covered area around selected nodes, do not explicitly characterize the charging station usage. As a result, these modeling approaches may fall short to support many real-world applications. On the other hand, while data-driven EV charging demand prediction models can be found in the literature (Amini et al. 2016, Almaghrebi et al. 2020, Wang et al. 2023, Zhang et al. 2023), the goals of these studies are different from CSLP, and don't consider the demand-supply coupling relationship when new charging stations are added to the network. These limitations highlight the need for incorporating explicit charging demand measurements into CSLP models to ensure their relevance and accuracy in practical scenarios.

To overcome these issues, in this manuscript we present a learning-based demand-supply-coupled optimization model for the charging station location problem, aiming to incorporate EV charging demand management into the planning of new charging infrastructures. We study a generic transportation network that has a certain number of pre-existing charging stations, and the planner has a pre-determined budget to build a fixed amount of new charging stations, choosing from a set of candidate locations. The choosing of new CS locations will change the offering of charging infrastructure supply in the neighborhood area, and subsequently, impact the EV charging demand that flows into the area, as well as the usage rate of the existing charging stations. As such, by "demand management", we refer to the use of new CS infrastructure planning as a system-level coordination. The goal is to maximize the utilization of all charging stations in the network, encompassing both newly constructed and pre-existing CSs. Compared with the literature, the proposed modeling approach does not require a pre-determined EV O-D demand matrix, but rather, learns from the field-collected charging dataset and explicitly predicts the charging demand of each charging station. Further, from the demand management perspective, the demand-supply

coupled interactions are modeled to account for the charging behavior variability associated with a new charging infrastructure offering. As such, instead of treating each charging station as an independent entity and naively selecting the candidate locations with the highest individual usage rates, we quantify the demand-supply coupled relationship at a system level and formulate CSLP optimization models with consideration of such interactions, with the goal of achieving the maximum utilization of all charging stations in the studied network.

Mathematically, a two-stage CSLP model is proposed to achieve these goals. In the first stage, instead of using the assigned traffic flow or service coverage area to represent charging demand, we use the charging usage rate as an explicit measurement, which is defined as the number of charging events per week. A gradient boosting-based ensemble learning model is developed to generate charging demand predictions. 15 features are defined, which cover spatial context information, weather information, charging infrastructure attributes, and traffic conditions in the surrounding area. Next, in the second stage, a demand-supply-coupled CSLP optimization model is developed with the objective of maximizing the total charging usage rates of both existing and newly selected CSs. The proposed model is solved with a gradient-based stochastic spatial search algorithm. To analyze the effectiveness of the proposed model, 6-year charging event data from Kansas City Missouri is used. We compared our modeling results with those from two benchmark models, namely, a greedy-based optimization model, and a neighbor swap-based optimization model. Results reveal that the developed stage one model is able to predict charging demand with R square values reaching 0.72 and 0.63 on the training and testing sets, respectively, indicating satisfactory performances in demand prediction. The stage two model is demonstrated to improve charging usage rates by 14% for the planned charging network, outperforming the results of both benchmark models.

The two-stage CSLP model we developed can directly assist agencies in determining optimal locations for new charging stations. In the early stages of electric vehicle adoption, planners primarily focused on installing stations in areas with limited infrastructure. Yet, as electric vehicles become more prevalent and charging infrastructure expands, planners now need to consider both the location and number of existing stations when planning new installations. This intricate balance between current and future stations adds a layer of complexity to the planning process. Our proposed model addresses this challenge by optimizing charging utilization rates for both existing and upcoming stations, presenting a strategic blueprint for this intermediate phase of charging infrastructure expansion. In a real-world test, our model proved invaluable, aiding the City of Kansas City, Missouri, in its plans for deploying new charging stations.

The following sections of this paper are structured as follows. Section 2 provides a summary of the literature review. Section 3 presents the formulation of the proposed two-stage model. The gradient-based stochastic spatial search algorithm is shown in Section 4 to solve the optimization model. Section 5 introduces a real-world instance to test the performance of the proposed model. Conclusions are summarized in Section 6.

2. LITERATURE REVIEW

A significant number of studies on CSLP have been found in the literature, with charging demand modeling being a key element of research. As reviewed by Kchaou-Boujelben (2021), according to the method of representing recharging demand, CSLP formulation can be mainly categorized into flow-based and node-based models.

Flow-based models assumed that EV drivers need to recharge their vehicles during the trip from origin to destination if they traveled for long distances, especially when the mileage exceeds the limited driving range. In this case, the charging demand was represented by a set of O-D trips

of EVs. Thus, the typical model setting was to optimize the CS location along these tips, with the objective of maximizing the number of EV drivers who can complete their journey from origin to destination and return without running out of battery. The common methods to describe charging demand coverage included flow capturing, flowing refueling, arc covering, path-segment, and battery state-of-charge (SOC) tracking (Kchaou-Boujelben 2021). Hodgson (1990) first proposed the flow-capturing location model, which assumed a trip was captured if at least one CS was placed along the path from origin to destination. Further, the flow refueling location model was introduced by taking EV driving range limitations into consideration (Kuby and Lim 2005). To address the issue of the high computational cost of flow refueling location models, a new arc covering-based formulation approach was presented, with the assumption that a trip of EV was covered as long as every arc consisting of the trip was covered by at least one CS (Capar et al. 2013). Unlike the arc covering method that decomposed trips into arcs, the path-segment-based coverage method divided each trip into a sequence of path segments that was composed of consecutive arcs (Yıldız et al. 2016). If a trip was considered to be captured, each path segment's length must not exceed the driving range of EVs, given that CSs were deployed at the starting node of each path segment. The battery SOC tracking approach was based on the idea of tracking the EV SOC value along every visited node, to ensure the remaining charge was always positive, otherwise recharging was triggered (Wang and Lin 2009). Although details may vary, flow-based research did not explicitly model the charging demand of an EV charging station. Instead, the covered EV traffic flow was frequently used as an optimization target, although the correlation between the charging usage rate of a CS and the bypassing traffic volume has not been clearly demonstrated. In addition, they usually assumed a known EV O-D demand matrix and mobility patterns of EV drivers, which is, however, not readily available in reality due to privacy concerns.

Unlike flow-based models that used EV trips to represent charging demand, node-based models, on the other hand, considered charging demand as a classical facility location modeling problem (Klose and Drexl 2005), i.e., charging demand generated at nodes in a planning area (Kchaou-Boujelben 2021). Such nodes may include many different types of POIs, where a group of EV drivers tend to prefer recharging, such as home, workplace, or restaurants. In terms of model formulation, existing node-based studies adopted approaches including location allocation, set covering, and maximal covering methods. Similar to the idea of the p-median problem (He et al. 2016), a location-allocation approach aimed to minimize the total system cost with all charging demands satisfied. The system costs typically included travel costs, which can be measured by the distances between EV users and the assigned CS, and fixed construction costs (Cui et al. 2019). The idea of the set covering approach was to cover every charging customer within the service area of CSs, which can be defined as the maximum driving time (Stephens-Romero et al. 2010) or maximum walking distance (Huang et al. 2016). Then, in some cases when not all EV users could be covered due to a limited budget, a maximal covering approach was adopted with the objective of maximizing the service coverage area of the planned CSs (Cavadas et al. 2015). However, existing node-based models typically used the size of the CS service area to represent the charging demand, and an explicit prediction of charging demand is not available.

The impacts of charging infrastructure supply on an EV driver's mobility behavior have been modeled in the literature. For example, considering charging costs and amenities at CSs, EV drivers were shown to deviate from the shortest path (Bernardo et al. 2016), as the charging customers wanted to maximize their utilities, and the route distance was only one of the several factors to consider. In another study, the pricing impact on the charging demand was modeled, where multiple charging service providers existed, and under the condition that all private

institutions aimed at maximizing their profit while satisfying the charging customers, the equilibrium of pricing was achieved (Guo et al. 2016). The study of EV drivers' decision-making behavior has also encompassed the form of bi-level flow-based models, which typically involve analyzing the decision process at the lower level and subsequently integrating it into the CS location optimization process of planners in the upper level (Chen et al. 2016, Wang et al. 2019). Additionally, as mentioned earlier in the context of node-based model approaches, charging stations were frequently built near points of interest such as restaurants or shopping malls, and studies have found that they play certain roles in EV driver's destination choice, i.e., where to travel to fulfill their needs (Nansai et al. 2001, Tuttle and Kockelman 2012). These studies suggest that the EV charging stations, as a new type of infrastructure, can be used as an effective system-level coordination to redistribute vehicular demand in the transportation network. However, the measurement of induced demand in response to new charging infrastructure should be carefully examined. To be specific, the deployment of a new CS may increase or decrease the charging demand of its neighboring existing CSs. For example, if the total charging demand remains stable, an increased number of charging stations may lead to a lower average usage rate at each charging station, whereas in some other cases, a denser charging infrastructure offering may make this area more attractive to EV drivers and encourages higher EV ownership, hence, the average usage rates at each charging station may increase. Modeling such demand-supply interactions becomes important to the formulation of CSLP.

A problem that is related to, but distinct from, our CSLP focus is the charging demand prediction, which has also been studied in the literature. For example, Almaghrebi et al. (2020) developed three machine learning methods to predict the energy consumed during the charging session. The utilized predictors included temporal information, user information, port number used

each session, fees policy, and land use category. A trip-chain-simulation-based charging demand prediction approach was proposed to estimate the number and spatial distribution of charging events and charging power (Zhang et al. 2023). Therein, over 60 EV behavior characteristic parameters were used covering the dimensions of time, spatial and energy, though the density of CSs was not considered. Amini et al. (2016) forecasted the charging demand of EV parking lots as time series data by an autoregressive integrated moving average (ARIMA) method. Then, a deep-learning-based Long Short-Term Memory (LSTM) neural network was developed to predict EV charging demand, which outperformed the traditional time series prediction models like ARIMA (Wang et al. 2023). Other data-driven charging demand prediction models can also be found in (Arias et al. 2017, Shahriar et al. 2021, Yi et al. 2022), among others. These models, while innovative, primarily rely on past data for predictions and may not account for shifts introduced by new infrastructures, a concept we refer to as the 'demand-supply coupling relationship.' Notably, the machine learning-based EV demand prediction models might not capture these evolving dynamics. The influence of charging infrastructure on an EV driver's mobility choices—like destination and route decisions—is paramount, and changes in these behaviors aren't necessarily reflected in models trained with existing EV charger data. Additionally, while pinpointing potential charging demand at candidate CS locations, these models don't provide strategies for station selection, underscoring the inadequacy of demand prediction alone in resolving the EV charger location challenge.

To summarize, the existing CSLP studies assumed the EV O-D demand and mobility patterns are readily available, which is not the case in reality, and the coupled demand-supply relationship results in the variability of O-D demand. Further, neither flow-based nor node-based CSLP models explicitly model EV charging demand, although they are frequently sought by the

stakeholders. To address these research gaps, a learning-based demand-supply-coupled optimization model is proposed. The stage one model explicitly predicts the charging demand of each CS with an ensemble learning approach, whereas the stage two model identifies the optimal CS location considering the demand-supply coupled interactions to maximize the charging usage rates of both existing CSs and selected charging locations.

3. PROBLEM FORMULATION

In this section, we present the mathematical formulation of the CSLP. As a convenient reference, the mathematical notations used in this manuscript are presented below.

i,j: Index of the CS location
w: Index of the week when charging events happen for one CS, $w \in W$
O: Input data set of the Stage one model
$a_{i,w}$: An array with the independent features of location i at week w
k: Index of the sub-regression trees in the ensemble learning model, k$\in K$
L: The objective function of the ensemble learning model
Θ: The set of all regression trees f_k in the ensemble learning model
f_k: The kth sub-regression tree
l: A training loss component
$\widehat{D}_{i,w}$: The observed charging demand of location i at week w
$D_{i,w}$: The predicted charging demand of location i at week w
Ω: A regularization component
γ: A penalty parameter to regulate the intricacy of the tree structure
T: The count of leaf nodes within a base tree
λ: A parameter to regulate the degree of regularization of f_k
Wt: The weight of leaf nodes
ln: The index of leaf nodes
N_{ln}: Total number of leaf nodes
obj: Objective function value of the Stage two model

N_e, N_c, N_d: Number of the existing CSs, candidate site locations, and new CSs to be deployed

$MAXN_z$, $MINN_z$: Maximum, and minimum number of selected new CSs in one zip code area

x_i: A binary decision variable, $x_i = 1$ when there exists or will be an EV CS at location i, otherwise $x_i = 0$. The same applies with x_j.

$NHCS_i, PRCP_w, TMP_w, WD_w, PT1{\sim}2_i, LDU1{\sim}7_i, AADT_w, TP_i$: Features to be introduced in Table 1

$CSNUM_i$: The overall count of CSs within the zip code where location i is located at

z_j^i: A binary decision variable, 1 if the zipcode of CS location j and i are the same, otherwise 0

s_i: Candidate CS location i in S

S^0, S: The initial solution set, and current solution set, of selected candidate CS locations

S', S^*: The updated solution set, and optimal solution set, of selected candidate CS locations

s_i^*: Candidate CS location i in S^*

$SC(s_i)$: Stochastic searching set for selected candidate location s_i

$NB(s_i)$, $NNB(s_i)$: The set of neighborhood and non-neighborhood candidate locations for s_i

$SUB_NNB(s_i)$: The set of randomly select candidate locations from $NNB(s_i)$

n_{rd}: The number of randomly select candidate locations in $SUB_NNB(s_i)$

α: A decay coefficient

E: Set of existing CS locations that has been built

e_i: An existing charging station at location i in E, $e \in E$

C: Set of not selected candidate locations

c_i: A non-selected CS location i in C

$IC(s_i)$: An interchange set for s_i

$CE_{i,w}$: The charging events from charging station i on week w

3.1 Overview of the two-stage optimization modeling framework

In this section we present a two-stage optimization model to identify the optimal locations for new charging stations, so that the charging usage rates of both existing and planned CSs are maximized. The data input includes charging event log data, travel demand model data and climate data. In

this section, we focus on the methodology development. The solution algorithm will be discussed in the next section, whereas a detailed description of the data is provided in the case study section.

Figure 1 below illustrates the proposed two-stage CSLP modeling framework. The purpose of the stage one model is to predict the charging demand at each candidate site, should it be chosen to build a charging station, with an explicit measurement that is represented by the weekly charging rate. The stage one model is formulated as a boosting-based ensemble learning approach, and it is assumed that the charging demand of a candidate site is dependent on an array of features that are associated with the candidate site. Then, the stage two model is developed to determine the optimal CSs location, accounting for the demand-supply coupled interactions, with the goal of maximizing the charging usage rates of both existing CSs and selected charging locations.

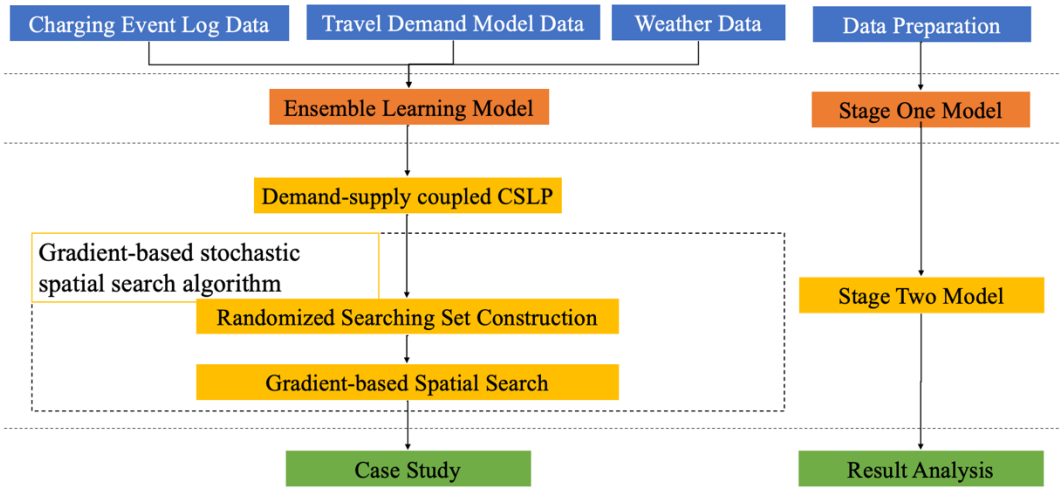


Figure 1 The workflow chart of the proposed model

The detailed modeling process will be explained in the following subsections. It should be noted that stage one learning model will be trained by the data that is collected from the field, which is a result of the current charging station layout and the corresponding EV charging behavior. However, once new charging stations are identified in stage two model, it leads to a new supply of the charging infrastructure, both in the total number of CSs and their spatial distribution, and subsequently, impacts the usage rates of the existing charging stations that have been built.

The demand-supply-coupled stage two model is designed to capture such dynamic interactions, and a gradient-based spatial search algorithm is designed to better estimate the marginal benefits of choosing a new candidate site at a network level, accounting for its impacts on the other charging stations.

3.2 Stage one ensemble learning model for CSLP explicit charging demand prediction

In this section, a boosting-based ensemble learning model is developed to predict the charging demand for CSLP, represented by the number of charging events each week, at each charging station. The input data of this model is $O = \{(a_{1,1}, \widehat{D}_{1,1}), (a_{1,2}, \widehat{D}_{1,2}), \dots, (a_{i,w}, \widehat{D}_{i,w})\}$, where $a_{i,w}$ is an array of features for charging station i during week w. The features that are included in $a_{i,w}$ are categorized into four groups, including spatial context information around the charging station, weather information, charging port type, and traffic information. A detailed definition of each feature will be given in Table 1 of the case study section. $\widehat{D}_{i,w}$ is the response variable, which is defined as the number of charging events at charging station i during week w. The target of the developed boosting-based ensemble learning model is to train optimal functions that minimize the prediction error between demand predictions and actual data that is obtained from the field.

A boosting-based ensemble learning model is developed and trained at this stage to improve the prediction performance. Among different kinds of machine learning techniques, ensemble learning models have been shown to be effective in improving the accuracy of predictions over traditional regression or classification models. This is achieved by training multiple sub-models on the same dataset and combining their results to obtain a more accurate prediction (Dong et al. 2020). Boosting is a popular ensemble learning technique that focuses on improving the accuracy of weak sub-models by iteratively adjusting the weights of the training

data based on the errors made by previous sub-models. The final model is a combination of all sub-models, and its predictions are based on a weighted average of the sub-models' predictions.

The formulation of the stage one charging demand prediction model is given from Eq. (1) to Eq.(5). The objective function, $L(\Theta)$, is formulated as Eq. (1), with the goal of minimizing the loss function l with a regularization term Ω to avoid overfitting.

$$Min\ L(\Theta) = \sum_{i=1}^{N_e} \sum_{w=1}^{W} l(D_{i,w}, \widehat{D}_{i,w}) + \sum_{k=1}^{K} \Omega(f_k) \quad (1)$$

Therein, Θ represents a set of K regression sub-trees that are denoted by f, as shown in Eq. (2). Each sub-tree, or regression model, in the set is designed to capture a unique aspect of the charging demand data and make a prediction based on a specific set of input variables. The output of each sub-tree is then combined to produce the final prediction of the model. the charging demand for a given set of input variables.

$$\Theta = \{f_1, f_2, \dots f_K\} \quad (2)$$

The loss function, denoted as l, is an important component in the charging demand prediction model. It is formulated as the squared error between the predicted charging demand $D_{i,w}$ and the actual value $\widehat{D}_{i,w}$ obtained from the field data, as shown in Eq. (3). This means that the performance of the prediction model is evaluated by how well it can minimize the difference between the predicted and actual values of the charging demand. By reducing this difference, the bias of the prediction model can be reduced, resulting in more accurate predictions.

$$l\left(D_{i,w}, \widehat{D}_{i,w}\right) = \left(D_{i,w} - \widehat{D}_{i,w}\right)^2 \quad (3)$$

The regularization term Ω is computed by Eq. (4), where T is the total number of leaf nodes in the base tree f_k. Parameter γ is designed to avoid tree structure from becoming too complex;

Wt is the weight of leaf nodes; the parameter λ controls the regularization level of f_k. The calculation of Wt will be discussed in the solution algorithm section.

$$\Omega(f_k) = \gamma * T + \frac{1}{2}\lambda * \|Wt\|^2 \quad (4)$$

Eq. (5) represents the final step in the proposed charging demand prediction model, where the predicted charging demand $D_{i,w}$ is obtained. Specifically, the predicted charging demand is calculated as the summation of the output of all base regression trees f_k, given the input feature vector $v_{i,w}$.

$$D_{i,w}(v_{i,w}) = \sum_{k=1}^{K} f_k(v_{i,w}) \quad (5)$$

With a sufficient number of sub-regression trees f_k built, the charging demand model in stage one will lead to a well-trained model that will then be integrated into the stage two model to generate an accurate charging demand value for each CS location. As the number of sub-trees increases, the accuracy and performance of the model will improve, therefore, the process of building and training sub-regression trees is crucial to the success of the charging demand model in stage one. Additionally, it is also important to select appropriate hyperparameters and regularization techniques to ensure that the model does not overfit or underfit the training data. The optimization process will be discussed in Section 4.

3.3 Stage two demand-supply coupled model for location selection optimization

The second stage model aims to optimize the location of charging stations by selecting the optimal candidate locations where all the charging infrastructure will be utilized the most. In a city network, we use N_e to denote the number of charging stations that have already been built. N_d represents the number of new EV CSs to be deployed, which will be chosen from a total of N_c candidate

locations. In other words, the objective is to choose N_d optimal locations, out of N_c candidate locations, so that the total charging demand, represented by the weekly charging rate of both planned and existing charging stations, can be maximized. The model formulation can be expressed as follows. Eq. (6) is the objective function and includes two terms. The first component is the summation of the charging demand of the newly planned CSs, whereas the second component is the summation of the charging demand of the existing CSs.

$$Max\ Obj\ (x_i) = \sum_{i=N_e+1}^{N_e+N_c} D_{i,w} * x_i + \sum_{i=1}^{N_e} D_{i,w} \quad (6)$$

In Eq. (6), $D_{i,w}$ represents the predicted charging demand at charging station candidate location i, during week w, and is calculated as the summation of the output of all base regression trees f_k from the stage one model, as shown in Eq. (7). This effectively connects the stage one model with stage two model, and is only made possible with a pre-trained stage one learning model with an explicit utilization rate for each charging station from the stage one model.

$$s.t.\ D_{i,w}(a_{i,w}) = \sum_{k=1}^{K} f_k(a_{i,w}) \quad (7)$$

An important input feature of $a_{i,w}$ in Eq. (7), $NHCS_i$, is the density of charging station in the neighborhood area that candidate location i is located at, and it is calculated via Eq. (8). $CSNUM_i$ is the number of total charging stations in neighborhood area, and $ARSZ_i$ is the size of the neighborhood area. It should be noted that $NHCS_i$ is an important input feature that reflects the charging demand-supply coupling relationship for EVs. On one hand, the goal of the stage two model is to identify the optimal candidate locations, to maximize the utilization of charging infrastructure, suggesting the EV infrastructure supply is determined by the charging demand. On the other hand, in the stage one model, the density of charging station, $NHCS_i$, is an important feature for charging demand prediction, meaning, the number of chargers in a neighborhood area

influences EV driver's decision of coming to the neighborhood and charging their vehicles while staying in this area. Together, they suggest CSLP is demand-supply coupled. As such, the $NHCS_i$ is used as a key variable that allows stage two model to iteratively improve the charging station layout. When a new CSLP solution is obtained, the density of charging station at each neighborhood area will need to be updated, which may result in a new charging demand for the CS in the region.

$$NHCS_i = CSNUM_i / ARSZ_i \tag{8}$$

Eq. (9) and Eq. (10) define the decision variable of the stage two model, x_i, which is a binary variable. x_i equals 1 if candidate location i is chosen to build a charging station, otherwise 0. Since the network includes a total of N_e existing charging stations, x_i equals 1 for those locations.

$$x_i = 1, i \in \{1,2, \dots, N_e\} \tag{9}$$

$$x_i = \begin{cases} 1 & if \ \ i \ is \ selected \\ 0 & Otherwise \end{cases}, i \in \{N_e + 1, N_e + 2, \dots, N_e + N_c\} \tag{10}$$

Eq. (11) defines another binary variable, z_j^i, that indicates the spatial relationship of two candidate locations, i and j. If they are located in the same neighborhood area, z_j^i takes a value of 1, otherwise 0.

$$z_j^i = \begin{cases} 1 & if \ i \ and \ j \ are \ in \ the \ same \ area \\ 0 & Otherwise \end{cases} \tag{11}$$

With x_i and z_j^i defined, we can proceed to calculate the $CSNUM_i$, the number of charging stations in the neighborhood area that candidate location i is located at. The mathematic form can be found in Eq. (12). Essentially, for each neighborhood area, it loops through all stations j to check if it is chosen to build charging stations, and if it is located in that neighborhood area.

$$CSNUM_i = \sum_{j=1}^{N_e+N_c} x_j * z_j^i \quad (12)$$

Eq. (13) is an equity constraint that specifies the number of new CSs deployed in any neighborhood area should be no less than a minimal number $MINN_z$, but no greater than $MAXN_z$.

$$MINN_z \leq \sum_{j=N_e+1}^{N_e+N_c} x_j * z_j^i \leq MAXN_z \quad (13)$$

And finally, in Eq. (14), the total number of newly deployed CSs in the city network cannot exceed N_d.

$$\sum_{i=N_e+1}^{N_e+N_c} x_i = N_d \quad (14)$$

4. SOLUTION ALGORITHM

In this section, the solution algorithms of both stage one and stage two models are presented. Therein, the stage one model is solved by an exact approach with explicit optima. The stage two model is solved with a proposed gradient-based stochastic spatial search algorithm.

4.1 Solution algorithm for stage one ensemble learning model

The stage one model in our EV charging infrastructure planning approach follows the concept of sequential fitting of weaker regression trees on the prediction residuals, proposed by Ke et al. (2017). This approach enables continuous reduction of the training loss by adding new prediction results until convergence. Furthermore, we show how to compute the second-order gradient and utilize this important information to optimize the objective function with exact solution formulation. This technique enhances the accuracy of the model by taking into account the curvature of the objective function and improving the convergence rate. As a result, our stage one model utilizes the benefits of sequential training and second-order gradient to effectively reduce the bias in the charging demand prediction model.

First, we substitute Eqs. (3) and (5) into the objective function in Eq. (1) and expand the square term. With some reorganization, Eq. (1) can be re-formulated as Eq. (15) below. At a current iteration K, the goal is to find a new sub-regression tree f_K, to iteratively reduce the prediction residuals.

$$Min\ L(\Theta) = \sum_{i=1}^{N_e}\sum_{w=1}^{W}\left(f_k(a_{i,w})^2 + 2\left(\sum_{k=1}^{K-1} f_k(a_{i,w}) - \widehat{D}_{i,w}\right) * f_k(a_{i,w})\right) + \Omega(f_K) + \sum_{i=1}^{N_e}\sum_{w=1}^{n} \widehat{D}^2{}_{i,w} + \sum_{k=1}^{K-1}\Omega(f_k) \quad (15)$$

In Eq. (15), we note the fourth term, $\sum_{i=1}^{N_e}\sum_{w=1}^{n} \widehat{D}^2{}_{i,w}$ is a constant, as $\widehat{D}_{i,w}$ is the actual utilization rate of EV charger that is obtained from the field data. Additionally, the last term, $\sum_{k=1}^{K-1}\Omega(f_k)$ is the regulation term associated with the trees that have been trained in the previous $K-1$ iterations. Hence, these two constant terms can be skipped, as they do not contribute to the problem solution. Eq. (15) can thus be simplified to Eq. (16) below.

$$Min\ L(\Theta) \cong \sum_{i=1}^{N_e}\sum_{w=1}^{W}\left(f_k(a_{i,w})^2 + 2\left(\sum_{k=1}^{K-1} f_k(a_{i,w}) - \widehat{D}_{i,w}\right) * f_k(a_{i,w})\right) + \Omega(f_K) \quad (16)$$

We then substitute Eq. (4) into Eq. (16), and expand the square term, $\|Wt\|^2$, which is the weight of leaf nodes. With some reorganization, Eq. (16) can be reformatted as Eq. (17) below.

$$Min\ L(\Theta) = \sum_{ln=1}^{N_{ln}}\left((Wt_{ln})^2 + 2\left(\sum_{k=1}^{K-1} f_k(a_{i,w}) - \widehat{D}_{i,w}\right) * Wt_{ln}\right) + \gamma * T + \frac{1}{2}\lambda * \sum_{ln=1}^{N_{ln}}(Wt_{ln})^2 = \sum_{ln=1}^{N_{ln}}\left((1+\frac{1}{2}\lambda)(Wt_{ln})^2 + 2\left(\sum_{k=1}^{K-1} f_k(a_{i,w}) - \widehat{D}_{i,w}\right) * Wt_{ln}\right) \quad (17)$$

After the reorganization, it can be found that in Eq. (17), the new objective function is in the form of a quadratic function of Wt_{ln}, where ln is the index of leaf nodes. The optimal solution can be obtained with the first derivative $\partial L(\Theta)/\partial Wt_{ln} = 0$, as shown in Eq. (18).

$$\frac{\partial L(\Theta)}{\partial Wt_{ln}} = \sum_{ln=1}^{N_{ln}} \left((2+\lambda) * Wt_{ln} + 2\left(\sum_{k=1}^{K-1} f_k(a_{i,w}) - \widehat{D}_{i,w}\right)\right) = 0 \quad (18)$$

Thus, for the new sub-regression tree f_K to be built in the current iteration, when the weight of leaf node ln, Wt_{ln}, takes the value of $Wt_{ln}^* = \frac{2\left(\sum_{k=1}^{K-1} f_k(a_w) - \widehat{D}_{i,w}\right)}{2+\lambda}$, this new sub-regression tree can reduce the prediction residuals to the maximum degree. The new prediction residuals can then be obtained via Eq. (19).

$$L(\Theta)^* = \frac{4*\left(\sum_{k=1}^{K-1} f_k(a_{i,w}) - \widehat{D}_{i,w}\right)^2}{2+\lambda} + \gamma T \quad (19)$$

4.2 Solution algorithm for stage two location selection model

In this section, a gradient-based stochastic spatial search algorithm is presented to solve the stage two location selection model. It consists of two components, namely, randomized searching set construction, and gradient-based spatial interchange. The solution framework of the stage two model is presented below. The basic idea is that starting from an initial CSLP solution S^0, in which S^0 is the set of chosen candidate locations s_i, with $S^0 = \{s_i | i \in \{N_e + 1, N_e + 2 \dots, N_e + N_c\}\ and\ x_i = 1\}$, construct a stochastic search set for each charging station s_i, and perform gradient-based spatial interchange to identify a candidate location with the highest charging demand improvement. This new candidate location will be used to swap out s_i in the previous solution, and finally, iteratively repeat this process until the stopping criteria is satisfied. The convergence error between two consecutive iterations is computed and compared to a pre-defined threshold err, to determine if the algorithm should terminate.

Framework: Gradient-based stochastic spatial search algorithm
Input: initial EV charging candidate location selection: $S^0 = \{s_i \| i \in \{N_e + 1, N_e + 2 \dots, N_e + N_c\}\ and\ x_i = 1\}$

Repeat

For s_i in current selection S where $i = N_e + 1$ to $N_e + N_c$ and $x_i = 1$, do

Construct stochastic searching set, $SC(s_i)$ (Component 1)

Perform gradient-based spatial interchange to generate new CSLP solution, S' (Component 2)

Until $Obj(S') - Obj(S) < err$

Output: optimal EV CS location selection plan: $S^* = \{s_i^* | i \in \{N_e + 1, N_e + 2 \dots, N_e + N_c\}\ and\ x_i = 1\}$

The first component is the stochastic searching set construction, which is described in detail below. For each chosen candidate location s_i, in the existing solution, its searching set $SC(s_i)$ is formed by including the candidate locations in its adjacent neighborhood areas (such as neighboring traffic analysis zones or zip code areas), $NB(s_i)$. To prevent the algorithm from being trapped in local minima, stochasticity is introduced by randomly selecting a set of candidate locations from the non-adjacent areas, denoted as $SUB_NNB(s_i)$, and adding them into $SC(s_i)$. A decay coefficient, α , is used to balance the exploration and exploitation. Initially, the number of non-adjacent candidate locations in set $SUB_NNB(s_i)$ is set to be relatively high, which encourages exploration. As the iterations progress, the number of non-adjacent candidate locations decreases, encouraging exploitation and leading to convergence towards a stable solution. The output of Component 1 is a set of candidate locations, which will then be used as the input data for Component 2, gradient-based spatial interchange.

Component 1: Stochastic searching set construction

Input:

A set of neighborhood candidate locations for the selected candidate location s_i with new charging station in current solution *S*: $NB(s_i)$

A set of non-neighborhood candidate locations for the selected candidate location s_i with new charging station in current solution S: $NNB(s_i)$

Randomly select n_{rd} number of candidate locations from $NNB(s_i)$, insert them into an empty set $SUB_NNB(s_i)$

Construct searching set for s_i: $SC(s_i) = NB(s_i) \cup SUB_NNB(s_i)$

Update n_{rd} with a decay coefficient, α, $n_{rd} = n_{rd} * \alpha$

Output: Searching set $SC(s_i)$

The second component is a gradient-based spatial interchange and is described as below. For a chosen candidate location s_i in the existing solution, a candidate location is firstly selected from the searching set and then added into the current solution set to replace s_i. If the total charging demand of the updated solution set is higher than before, such interchange between s_i and the candidate location is considered favorable and will be added to the potential interchange set. The marginal improvement, $\nabla obj(x_i) \geq 0$, is computed with Eqs. (6)-(12). After every candidate location in the searching set has been iterated, the optimal interchange in the potential interchange set will be used to update the current solution set, which is expected to iteratively improve the value of the objective function.

Component 2: Gradient-based spatial interchange

Input:

Set of currently selected candidate locations: $S = \{s_i | i \in \{N_e + 1, N_e + 2 \dots, N_e + N_c\} \; and \; x_i = 1\}$

Set of existing CS locations that has been built: $E = \{e_i | i \in \{1,2 \dots, N_e\}\}$

Set of not selected candidate sites: $C = \{c_i | i \in \{N_e + 1, N_e + 2 \dots, N_e + N_c\} \; and \; x_i = 0\}$

Objective function value with S and E, $obj(x_E, x_S, x_C)$, computed with Eq. (6)

The searching set for s_i: $SC(s_i) = \{sc_p | \; p \in \{1,2 \dots, n_{rd}\} \; and \; sc_p \in C\}$ from Component 1

Interchange set for s_i: $IC(s_i) = \emptyset$

Repeat for each candidate location, sc_p, from $SC(s_i)$:

Set $x_i = 0$, and $x_{sc_p} = 1$

Compute the gradient in respect to x_i:

Let $S' = S - s_i + sc_p$, $C' = \mathrm{C} + s_i - sc_p$,

Compute new objective function value $obj(x_E, x_{S'}, x_{C'})$, with Eqs. (6)-(12)

$\nabla obj(x_i) = obj(x_E, x_{S'}, x_{C'}) - obj(x_E, x_S, x_C)$

If $\nabla obj(x_i) > 0$: $IC(s_i) = IC(s_i) \cup sc_p$

Select the new charging station location from $IC(s_i)$ that maximize $\nabla obj(x_i)$

Output: Updated solution set: $S' = S - s_i + sc_p$, $C' = \mathrm{C} + s_i - sc_p$

5. CASE STUDY

In this section, we evaluate the effectiveness of our proposed two-stage model using a real-world dataset from Kansas City, Missouri (KCMO). Firstly, we will provide an overview of the dataset, including some basic characteristics. Next, we will define the features to be used in the stage one model. Finally, we present the numerical analysis results for stage one and stage two models, respectively.

5.1 Data description and feature definition

This research uses a multi-source dataset collected in KCMO, which includes the charging event log data, travel demand model data, and climate data. A detailed description of each data source is provided below. In total, 67,576 data samples are extracted, and 15 features are defined in Table 1. These features can be categorized into four main categories: 1) spatial context information, 2) weather information, 3) charger type and 4) traffic information. A detailed definition of these features is presented in the subsections below. Notably, some variables are continuous numerical data, while others are categorical data.

Table 1 Definition of 15 Features and Response Variable

Category	Notation	Definition and Unit	Data Type
Response Variable	$D_{i,w}$	The number of charging events of CS i on week w	Numerical
Spatial Context Information	$NHCS_i$	The density of charging stations in the same zip code area of CS location i (#CSs/sq mile)	Numerical
	$LDU1_i$	Land use type: if CS location i is in an institutional area	Binary
	$LDU2_i$	Land use type: if CS location i is in a transportation area	Binary
	$LDU3_i$	Land use type: if CS location i is in a commercial area	Binary
	$LDU4_i$	Land use type: if CS location i is in a residential area	Binary
	$LDU5_i$	Land use type: if CS location i is in a recreational area	Binary
	$LDU6_i$	Land use type: if CS location i is in a vacant area	Binary
	$LDU7_i$	Land use type: if CS location i is in an industrial area	Binary
Weather Information	$PRCP_w$	Weekly precipitation of week w (mm)	Numerical
	TMP_w	Weekly average temperature of week w $(°C)$	Numerical
	WD_w	Weekly average wind speed of week w (m/sec)	Numerical
Charger Type	$PT1_i$	Port type of DC fast charger for CS location i	Binary
	$PT2_i$	Port type of level 2 charger for CS location i	Binary
Traffic Information	$AADT_i$	Annual average daily traffic on CS location i's nearby roads	Numerical
	TP_i	Trip production of the TAZ where CS location i is located	Numerical

With these 15 features defined, $a_{i,w}$, representing the array of features for charging station i during week w as defined in the stage one model, can be rewritten with Eq. (20) below.

$$a_{i,w} = [NHCS_i, PRCP_w, TMP_w, WD_w, PT1{\sim}2_i, LDU1{\sim}7_i, AADT_w, TP_i] \quad (20)$$

5.1.1 Charging event log data

The charging event log dataset comprises 22,0231 charging records, collected from 444 public charging stations in KCMO (i.e., $N_e = 444$), from January 2014 to December 2019. The spatial distribution of existing CSs is shown in Figure 2. The concentration of charging stations (CSs) is notably prominent in the downtown area, where most of the high-demand CSs are also observed.

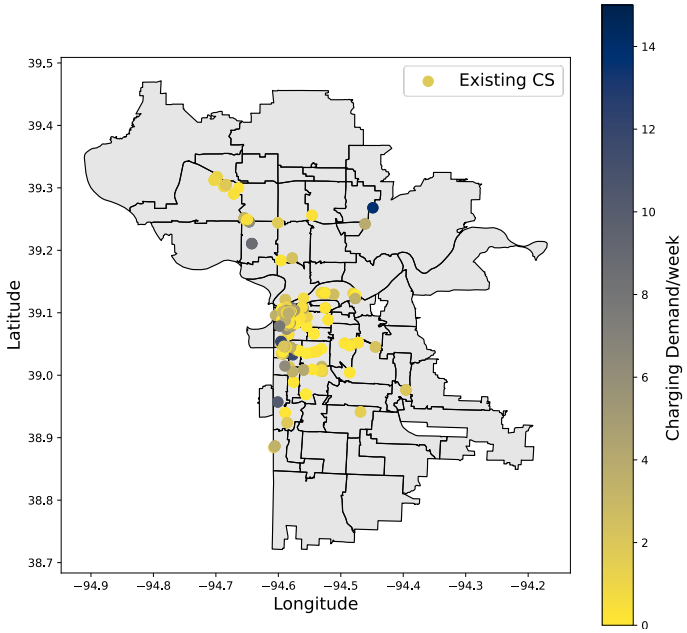


Figure 2 The spatial distribution of existing CSs in Kansas City, MO

For the charging event dataset, the available attributes include start date, end date, latitude, longitude, plug type, and postal code of a charging event. With this information, we can derive the response variable and features including spatial context information and port type. The response variable is the weekly charging rate, which is shown in Eq. (21), where $count(CE_{i,w})$ denotes the total number of charging events from charging station i on week w. Its distribution is shown in Figure 3. It can be observed that for each CS, the number of charging events is mostly under 10 times per week, with a longtail where the maximum reaches 58.

$$D_{i,w} = count(CE_{i,w}) \tag{21}$$

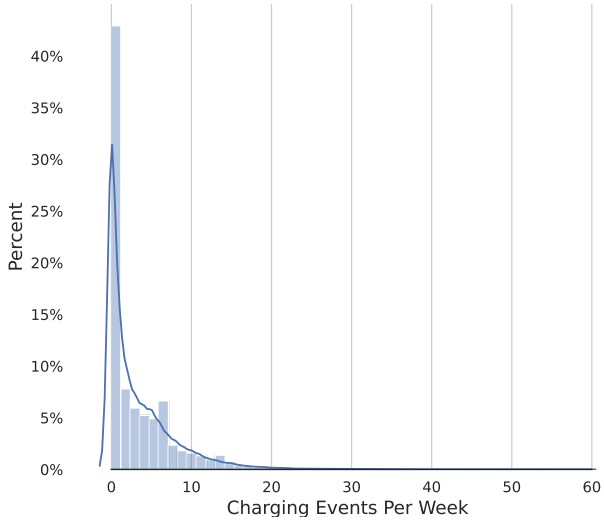
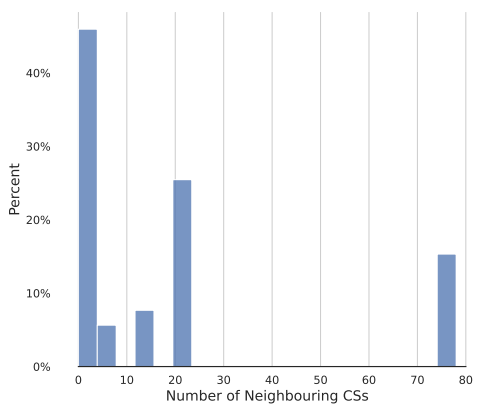
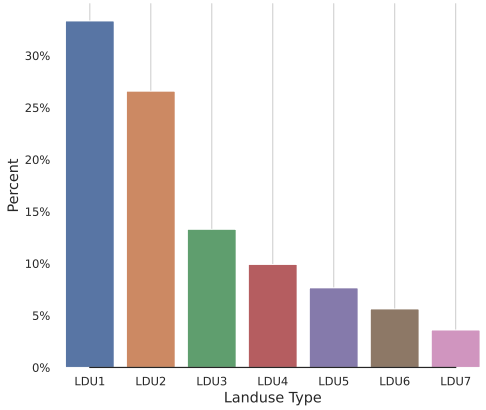


Figure 3 The distribution of response variable, charging demand

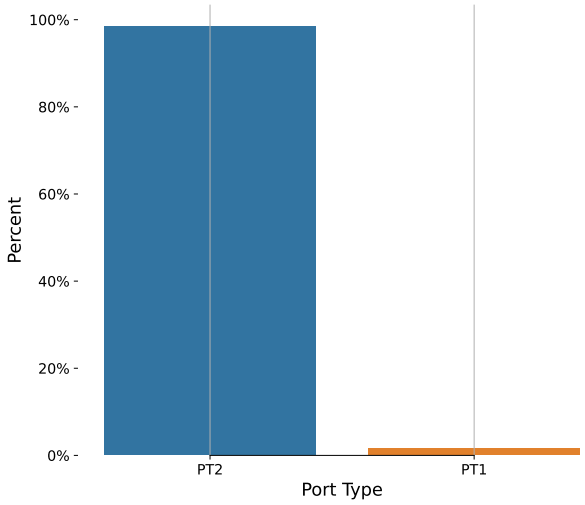
When it comes to relevant features, the number of neighboring existing CSs, $NHCS_i$, is defined as the density of charging stations (number of CSs per sq mile) in the same zip code area of CS location i, as shown in Eq. (8). Of all 444 stations, the distribution of $NHCS$ is given in Figure 4-(a), where we can find three clusters. Such clusters denote that the density of CSs may be below 10, around 20, or over 70 per sq mile. $LDU1{\sim}7$ are the land use type of the parcel where the CS i is located. In Figure 4-(b), the distribution of seven land use types of 444 CS locations is presented. Most of the CSs are located in $LDU1$ and $LDU2$, i.e., institutional areas such as offices and schools, transportation areas such as stations and airports. As Figure 4-(c) illustrates, the majority of charging stations are equipped with the level 2 chargers ($PT2_i$), whereas $PT1_i$ is the fast charger and accounts for only about 5% of the charging stations.



(a) Number of Neighboring CSs



(b) Land use Type



(c) Port Type

Figure 4 The distribution response variables

5.1.2 Travel demand model data

The travel demand model data is sourced from Mid-America Regional Council (MARC), which is the Metropolitan Planning Organization (MPO) for the bi-state Kansas City region. The data encompasses a daily origin-destination (O-D) demand matrix and a road network of Kansas City, featuring assigned peak hour traffic volume. Kansas City is divided into 2,510 Traffic Analysis Zones (TAZs) with a total of 10,533 nodes and 24,601 links. The trip production value for each

TAZ can be calculated through summarizing the OD demand matrix by rows. Therefore, the feature TP_i, representing the trip production of the TAZ where CS i is located, can be calculated using Eq. (22), where $TPSUM_i$ is the summation of trip production generated within the zip code area where CS i is located; $ARSZ_i$ denotes the area size of the zip code zone of CS i. It should be noted that the obtained O-D demand matrix is for ICE vehicles and EV combined, and an EV-only O-D demand is not available nor needed for this case study.

$$TP_i = TPSUM_i / ARSZ_i \tag{22}$$

$AADT_i$, the annual average daily traffic on CS location i's nearby roads, is calculated by Eq. (23), where $AADTSUM_i$ represents the total daily traffic volume summed across the roadway segments within the zip code zone where CS location i is situated. Though TP_i and $AADT_i$ are both traffic-related features, it should be noted that TP_i focuses on the number of vehicles originating from a particular zone, while $AADT_i$ includes both the local traffic and the bypass traffic.

$$AADT_i = AADTSUM_i / ARSZ_i \tag{23}$$

5.1.3 Climate data

The climate data is retrieved from NCEI via open access API, and includes weather station location, record date, average wind speed in m/sec, precipitation in mm, and the average temperature in $°C$. For the weather-related features, the weekly average temperature of week w, TMP_w can be calculated by Eq. (24), where $Tavg_w$ is the average daily temperature of date in week w. Similar calculation is also performed to get WD_w, the weekly average wind speed of week w.

$$TMP_w = sum(Tavg_w)/7 \tag{24}$$

The weekly precipitation of week w, $PRCP_w$, can be derived by Eq.(25), where $Prcp_w$ is the precipitation value of each day in week w.

$$PRCP_w = sum(Prcp_w) \quad (25)$$

After data processing, the total number of data records is 67,576, each including the 15 defined features and the response variable. Such dataset is randomly divided into a training subset which has 80% of the data (54,061 samples), and a testing subset with 20% data (13,515 samples).

5.2 Test scenario setup

To validate the two-stage model, 5 points of interests are pre-identified for each zip code area and used as candidate CS locations. The studied area includes a total of 71 zip codes, so in the end, we have N_c=355 (i.e., 5*71 zip code areas). N_d is set to be 50, i.e., 50 new charging stations will be selected from 355 candidate locations, with the goal of maximizing the charging utilization of both 444 existing (i.e., N_e) and 50 new CSs to be built (i.e., N_d). Figure 5 illustrates the spatial distribution of the 355 candidate locations.

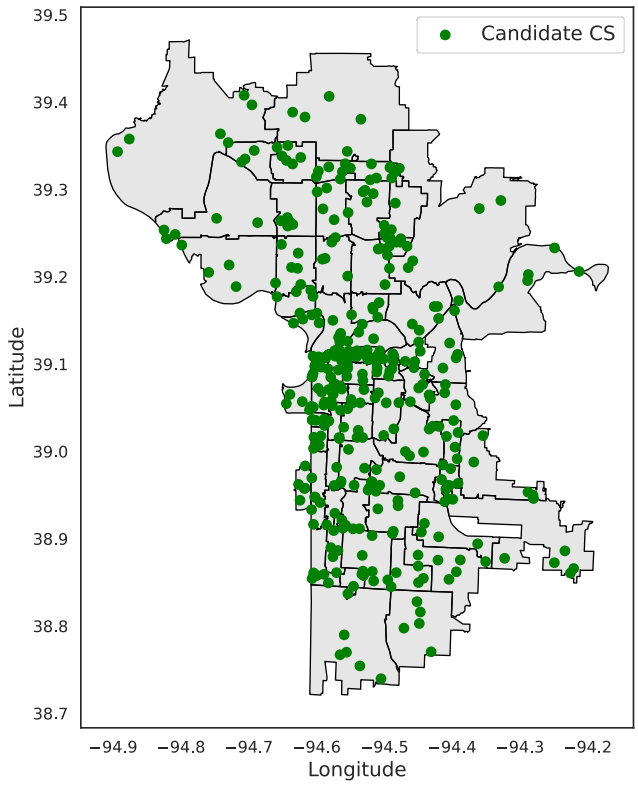


Figure 5 The distribution of 355 candidate CS locations

The other parameters are set as follows. The maximal and minimal number of new-built CSs in a single zip area is set to be 3 and 0 (i.e., $MAXN_z = 3\ and\ MINN_z = 0$). In the solution process, the initial number of candidate locations selected from $NNB(c_m)$, i.e., n_{rd}, is set to be 15. The convergence criteria between two iterations to terminate searching (i.e., err) is set to 0.01. The numerical analysis is implemented in a Python environment on a desktop computer featuring an Intel (R) Xeon (TM) W-2295 CPU 18 Core@ 3.00 GHz and 128 GB DDR4 RAM.

5.3 Stage one ensemble learning model training results

With the scenario setting as described above, a total of 51,896 sub-regression trees are built. The prediction performance of the model on the training and testing dataset is measured by R square, root mean squared residuals (RMSE), and mean absolute error (MAE). As shown in Figure 6, R square values are 0.72 and 0.63, RMSE are 2.38 and 2.77, and MAE are 1.62 and 1.85, on the

training and testing subsets, respectively. Compared with the charging demand prediction models from the literature, whose training and testing r square values were shown to be between 0.567 and 0.519 (Almaghrebi et al. 2020), the proposed model is shown to give satisfactory performance in generating accurate charging demand predictions.

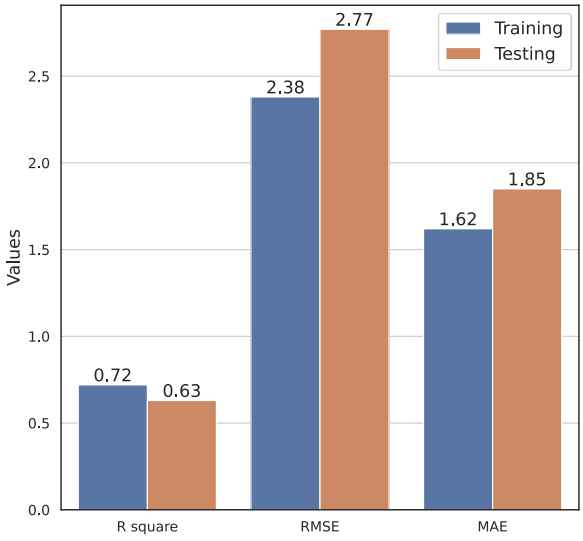


Figure 6 The prediction performance on training and testing sets

To further validate the prediction performance of the stage one model, three other representative prediction models are implemented using the same data set for comparison, namely Multiple Linear Regression (MLP), K-Nearest Neighbors (KNN), and Neural Networks (NN). The comparison results on the testing set are given in Table 2. It can be observed that the proposed model outperforms the other prediction models with the highest R square value and the lowest RMSE and MAE. Thus, the stage one model could give relatively accurate prediction results because of its ability to utilize the boosting techniques to combine weak learners into a strong predictive model.

Table 2 Performance comparison among four models

Models	R square	RMSE	MAE
Proposed	0.63	2.77	1.85
MLP	0.20	4.09	2.92
KNN	0.29	4.13	2.42
NN	0.40	3.57	2.51

To assess the influence of the utilized features on charging demand, the importance of each feature is quantified by the Shapley Additive Explanation (SHAP) method, which produces consistent output by averaging all possible orderings of input feature permutations (Lundberg et al. 2018). The top ten features with the largest importance value are given in Figure 7, with X-axis being the mean SHAP value that measures the average impact of a specific feature on modeling output, and Y-axis being the features. It is found that the two most important predictors are both traffic-related, nearby traffic $AADT_i$ and trip production TP_i. The number of neighboring CSs, $NHCS_i$ ranks fourth, which validates the demand-supply coupled relationship, as charging demand is shown to be affected by the density of CSs in the same neighborhood.

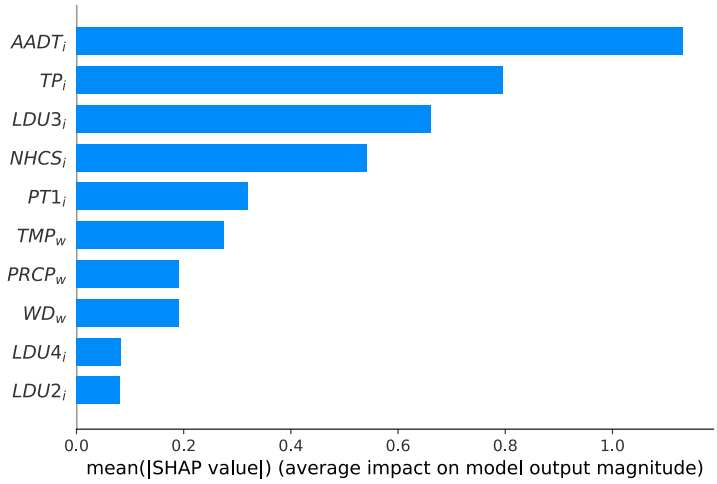


Figure 7 The mean SHAP value for the 10 most important features

5.4 Stage two location selection model results

For stage 2 model validation purpose, we fix the value of 14 features in Eq. (7), and only change the charging station density, $NHCS_i$, so as to better focus on its impact of supply on the usage rate of new and existing charging stations. Regarding the remaining features, they have been drawn from a data sample characterized by the lowest prediction MAE value. This selection is grounded in the anticipation that these specific parameter values will result in the most precise charging demand predictions. Furthermore, this particular sample is reflective of the prevailing conditions and trends commonly observed in KCMO, encompassing factors such as port type, traffic information, and weather data. Specifically, we set $LDU3_i = 1$, i.e., commercial area, and thus the value of other land use related variables become zero. The other variables are set as the following: $PRCP_w = 5.60\ (mm)$, $TMP_w = 6.52\ (°C)$, $WD_w = 5.11$ (m/sec), $PT1_i = 0$, $PT2_i = 1$, $AADT_i = 4821$, and $TP_i = 5399$.

The convergence of the stage two model is shown in Figure 8 below, with X-axis representing the number of iterations and Y-axis corresponding to the objective function value. It can be found that as the iteration goes, the objective function value, predicted weekly charging rates, increases rapidly first, then the growth slows, and the result becomes stable. The EV charging utilization is shown to have improved from the initial solution of 1,278 to the final optimal solution of 1,459 after eight iterations. The Figure also shows the objective function does not change much after the third iteration, indicating the proposed model has a fast convergence property.

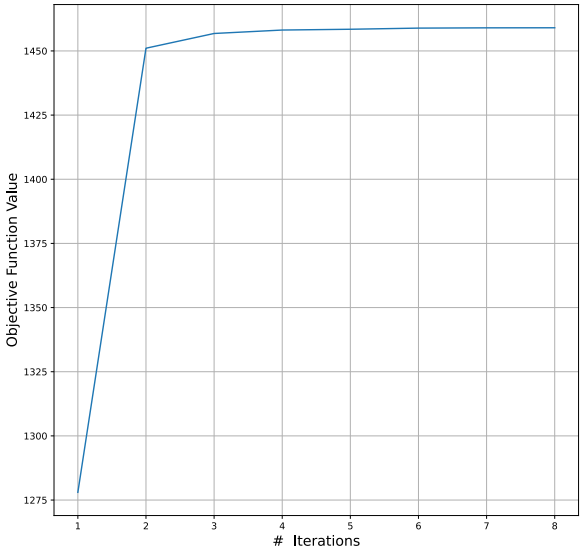
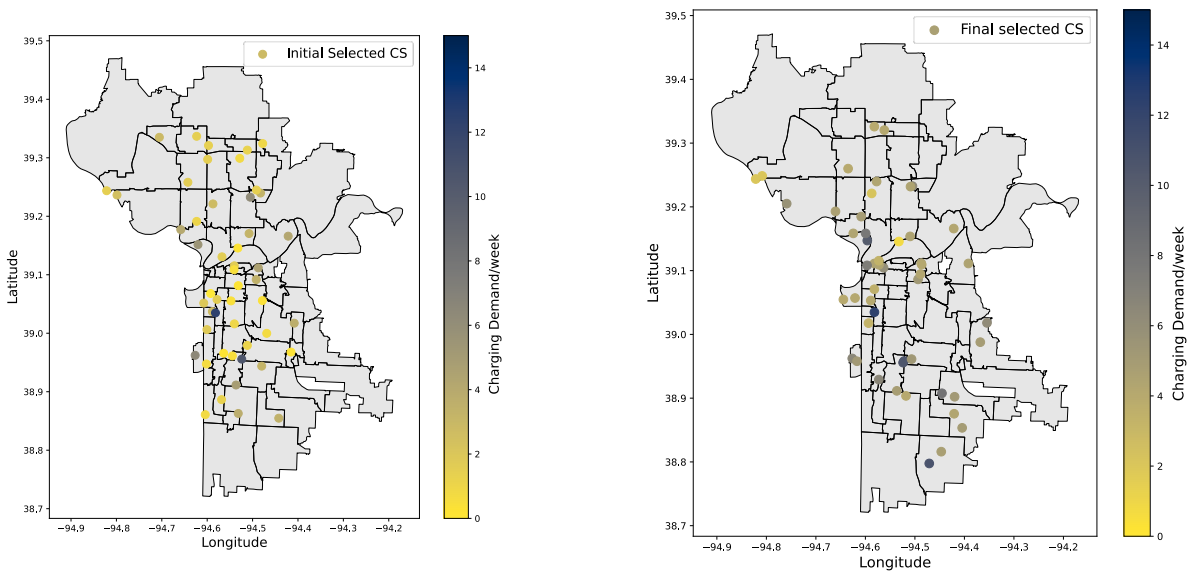


Figure 8 Solution iteration process

Then, the spatial distribution of CSs in the initial and final solution is shown in Figure 9 (a) and (b), respectively. Each dot represents the location of an EV charger. It can be found that the fifty charging stations in the initial solution are evenly distributed on the map, with a maximum of one charging station in each zip code area. However, in the optimal solution, the finally selected 50 CSs are more concentrated in certain zip code areas. For example, one zip code area has three locations selected, and eighteen zip code areas have two locations selected. Eleven zip code areas are assigned with one charging stations. From Figure 9-(b), it is evident that zip code areas with multiple charging stations are predominantly concentrated in the downtown area, reflecting the higher demand for EV charging in comparison to other regions.



(a) Initially selected CSs

(b) Finally selected CSs

Figure 9 Distribution of initial and final selected CSs with their utilizations

We use the color of the dots in Figure 9 to illustrate the changing demand. Yellow color indicates a low demand, whereas a darker color stands for a higher EV charging utilization. It can be found that in the initial solution in Figure 9 (a), most locations are colored with lighter yellow, meaning their utilizations are relatively low. On the other hand, the color of CSs in Figure 9 (b) is much darker, indicating the charging demand of the finally selected CSs are higher than the initial solution, and that the proposed two-stage model is able to identify EV charging station locations that increase the charging usage rates in the network.

5.5 Comparison with two benchmark models

To test the effectiveness of the proposed solution algorithm, we compared our modeling results (indicated as "Proposed Method") with those from two benchmark approaches. Benchmark 1 employs a Greedy-Based Method. This method opts for candidate locations exhibiting the highest predicted charging demand per the stage one model. In other words, Benchmark 1

algorithm recreates a scenario where only charging station demand prediction is accounted for, sidelining the demand-supply coupling relationship, such as Almaghrebi et al. (2020) and Zhang et al. (2023). Benchmark 2 introduces the Neighbor-Swap-Based Method. This technique addresses the stage two model by limiting its selection updates to neighboring areas and excluding non-neighboring zones. Previous studies like those by Guo et al. (2018) and Schiffer and Walther (2017) had proposed adaptive large neighborhood search (ALNS) algorithms to enhance the objective function of the CSLP model. Their strategies primarily revolved around inspecting the neighborhood using various destroy and repair maneuvers. However, these methodologies confined their spatial search set to the neighborhood area, eschewing stochastic inputs from non-neighboring zones. Such an approach could skew the final solution based on its initial configuration. The result comparison is presented in Figure 10, in which we also display the separate utilized rates for the existing CSs, newly selected CSs, and all CSs combined. Also, the initial solution serves as a baseline, generated by randomly selecting 50 new charging stations from the 355 candidate locations.

Figure 10 suggests that for the 444 existing charging stations in Kansas City, their charging usage rates are impacted by the strategies of selecting new charging stations. The proposed method and the two benchmark approaches generate a charging usage rate of 1,195, 1,163, and 1,195 per week, respectively. Compared with the original utilization, which is 1,153 per week, the development of new charging stations increased the usage frequency of existing charging stations, by 3.5%, 0.9%, and 3.5%, respectively. The proposed method is shown to increase the utilization rates of the existing charging stations, and the effectiveness is similar to the second benchmark approach, whereas outperforms the first benchmark approach.

Next, we examine the utilization rates of the 50 newly developed charging stations. Compared with the initial solution that has a utilization rate of 125 per week, these three methods can all significantly increase charging rates, with benchmark approach one achieving the best performance. The final charging rates are 264, 266, and 256 for the proposed method, and the two benchmark approaches, respectively. If we only use these newly developed charging stations as the study target, benchmark approach one is shown to generate the best results, outperforming the proposed method and the second benchmark approach.

However, when all charging stations are combined (444 existing charging stations plus 50 selected new charging stations), the proposed method is shown to generate the highest usage rate, at 1,459 charging events per week, which is significantly higher than the initial solution (1,278), and outperforms the two benchmark approaches (at 1,429 and 1,451, respectively). In other words, the proposed method is able to identify the optimal charging locations in a way to better serve the overall charging demand in a network, not only for the newly selected charging stations but also for the existing charging stations that have been built. Also, the proposed solution algorithm could further improve the total charging demand by introducing stochasticity in the searching process. The results of the proposed method can improve charging usage by 14% over the initial solution, and outperforms the two benchmark approaches by 2% and 0.5%, respectively.

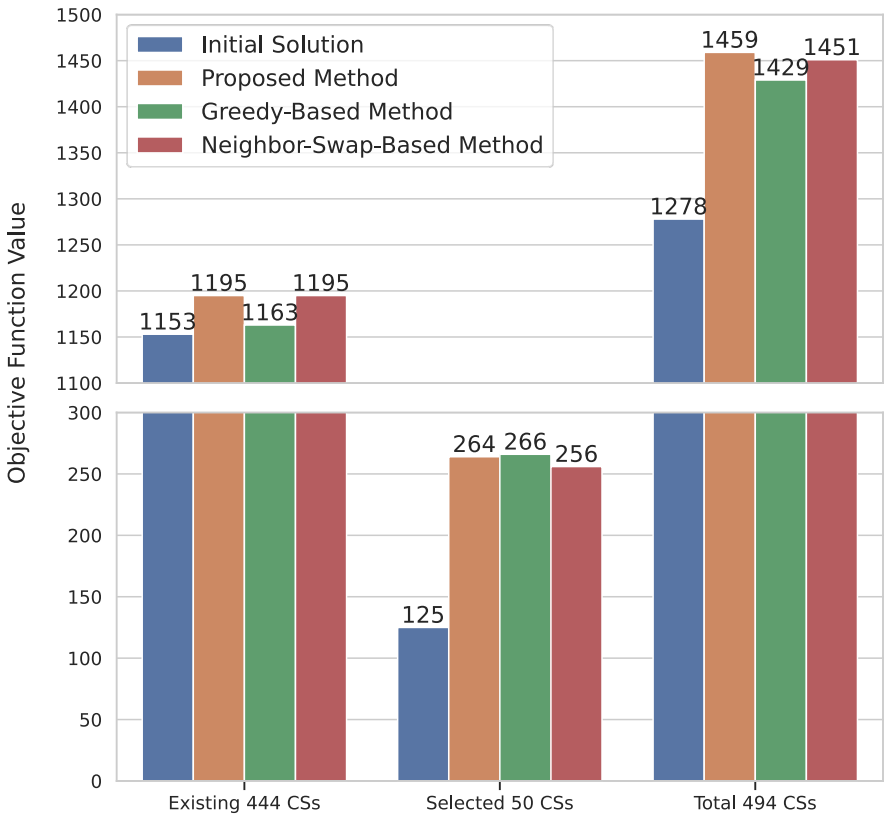


Figure 10 Solution Results

Next, to validate the performance of the proposed method under various parameter settings, we conduct a sensitivity analysis on the number of new charging stations, and the results are shown in Figure 11. We set the number of new charging stations to range from 30 to 110, with an interval of 20. It can be observed that the usage rate generated by the proposed method varies from 1370 to 1636 charging events per week. In all cases, it outperforms the initial solution and the two benchmark approaches. This indicates that the performance of our method is robust regardless of the number of new charging stations to be deployed. Notably, the gap between our method and the first benchmark, Greedy-Based Method, widens as the number of new charging stations increases. This can be attributed to the fact that the interplay between existing and newly selected charging stations intensify as the number of new charging stations grows.

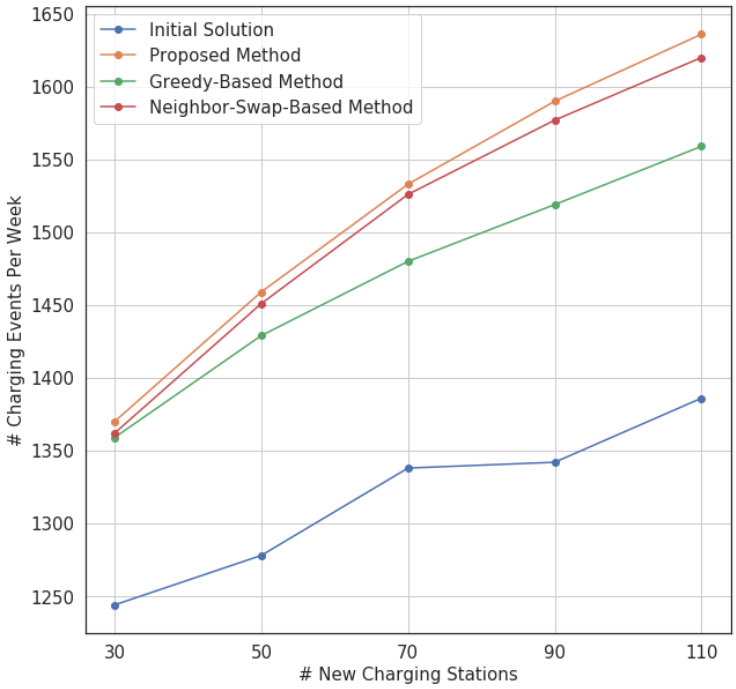


Figure 11 Sensitivity Analysis of Number of New Charging Stations

6. CONCLUSION

In this manuscript, a learning-based demand-supply-coupled optimization model for the charging station location problem is developed, aiming to incorporate electric vehicle charging demand management into the planning of charging infrastructures. This CSLP is formulated as a two-stage model. In stage one, a gradient boosting-based learning model is developed to predict the charging demand of a charging station based on 15 defined features. Next, in stage two, a demand-supply-coupled CSLP model is developed with the objective of maximizing the total charging usage rates of both existing and newly selected charging stations. The proposed model is solved using a gradient-based stochastic spatial search algorithm.

A case study using multi-source real-world data from Kansas City, Missouri is performed to test the effectiveness of the proposed model. The used dataset includes a total number of 22,0231 charging records, collected at 444 public charging stations in a 6-year span. The travel demand

model data and climate data are also retrieved for research purposes. Based on these multi-source data, a total of 15 features are extracted. Results show that the stage one model generates a satisfactory charging demand prediction result, with R square values reaching 0.72 and 0.63 on the training and testing datasets, respectively. The stage two model with the proposed solution algorithm is demonstrated to improve charging usage by 14%, and outperform the results of two benchmark approaches, including a greedy-based method, and a neighbor swap-based method. The results of this research are poised to guide agencies in identifying optimal locations for new charging stations. By maximizing utilization for both current and future stations, the model strategically optimizes placement during periods of infrastructure growth.

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8. AUTHOR CONTRIBUTIONS

The authors confirm their contribution to the paper as follows: study conception and design: Yang Song, Xianbiao Hu; data process: Yang Song; analysis and interpretation of results: Yang Song; draft manuscript preparation: Yang Song, Xianbiao Hu. All authors reviewed the results and approved the final version of the manuscript.

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